PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	More than just staffing? Assessing evidence on the complex interplay among nurse staffing, other features of organizational context and resident outcomes in long-term care: a systematic review protocol
AUTHORS	Choroschun, Katharina; Kennedy, Megan; Hoben, Matthias

VERSION 1 – REVIEW

REVIEWER	Park, Claire Center for Econometric Optimization in the Nursing Workforce
REVIEW RETURNED	15-Mar-2022

GENERAL COMMENTS	To authors, Thank you for your hard work. I did my best to give you constructive comments (please see below). I hope that my input is helpful for you to improve the manuscript.

REVIEWER	Mueller de Magalhaes, Ana Maria Universidade Federal do Rio Grande do Sul, Nursing School
REVIEW RETURNED	15-Mar-2022

GENERAL COMMENTS	The systematic review protocol addresses a relevant theme about interactions among organizational context factors and nursing staffing, and how these are associated with resident outcomes in long-term care. The study protocol is comprehensively described and allow to replicate the research strategy. The protocol is registered to the PROSPERO 2021 (CRD42021272671) and report an ongoing study. Minor revision According to the policies of the BMJ Open the dates of the study should be included in the manuscript (start of the registration; the first search strategy; the planned date to conducted the search strategy and data extraction)

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

Comment	Response

Protocol papers should report planned or ongoing studies. The dates of the study should be included in the manuscript.	Thank you for raising this issue. We updated this information in the methods and analysis section on p.3 lines 133-135.
Title: More than just staffing? Assessing evidence on the complex interplay among nurse staffing, other features of work environments organizational context and resident outcomes in long-term care – a systematic review protocol	We agree with the reviewer's suggestion. Since we consider staffing to be part of organizational context, we have adapted the title (p.1 lines 1-2).
These studies have been conducted several times in acute care settings. Is there any particular reason to do this type of research in long term care setting? What's that?	Thank you very much for these thoughtful considerations. We agree that the evidence on the association between staffing and resident outcomes is more robust in acute care than in LTC. However, acute care settings differ from LTC in various important ways. More importantly, this review does not propose to assess the direct association of staffing and resident outcomes (which is what most studies and literature reviews – in acute care, as well as, in LTC – have assessed). Our focus is the interaction between staffing and other contextual factors and the influence of the complex interplay among these factors on resident outcomes. To the best of our knowledge, this has not been done comprehensively in acute care either, let alone in LTC. We tried to clarify this in the abstract (p.1 lines 14-19) and in the introduction (p.2 lines 64-69).
For quantitative research studies and systematically conducted reviews: Why are you reviewing only these kinds of papers?	We appreciate the opportunity to clarify the selected study designs. Our research question focuses on what we know about the interactions between nurse staffing and organizational context in LTC homes, and the effects of these interactions on LTC resident outcomes. Interaction effects and their associations with other outcomes are statistical effects and statistical effects are not assessed in qualitative studies. Therefore, this review will only include quantitative studies. We elaborated on this in the abstract by describing that we are specifically interested in statistically modeled interactions (p.1 lines 25-29).
Indeed, will LTC nursing shortage only affect quality of care? Try to think about this. Why would not mainstream nursing workforce research intensively address nurse staffing	Thank you again for raising this important issue. We agree that cost-effectiveness and cost-efficiency of the nursing workforce are of great interest to policy makers. However, we

and cost of care? Nursing workforce policy makers may much more need/want the evidence of the cost-effectiveness and cost-efficiency of the nursing workforce.	do think that quality of care outcomes (i.e., the residents' health and well-being) cannot be sacrificed for reasons related to cost-effectiveness/-efficiency. Since this review cannot focus on all possible outcomes, we decided to focus on those that we considered highest priority: the residents' heath and well-being (quality of care).
Non-linear concept is very important in the current/future nursing workforce studies. The issue is that it is NOT your original idea. I strongly recommend you to conduct more literature reviews to support the point (non-linear associations, including why it is important) and respect the researchers by citing their articles. Also, please provide more than two rationales on why this sentence supports your argument (inconsistent evidence in LTC).	Thank you again for raising this important issue. We apologize for not having been clear enough. Our systematic review will include all studies that report interactions and their association with quality of care. This includes studies that assume (and model) a linear relationship, as well as, studies that use nonlinear models. We consider whether the relationship is linear or non-linear to be an empirical question, which we may be able to answer in our review. With our results, we hope to contribute to a better understanding of these complex relationships.
Be consistent, please: A, B, and C A, B and C Choose one and then revise it throughout the paper, please.	Thank you for mentioning this. We addressed this throughout the manuscript.
How different is "organizational context" from the structural variables? The structural variables may also influence the level of quality of care in LTC in a significant way. Why should "organizational context" be more importantly addressed than the structural variables?	Thank you again for raising this important issue. We apologize for not having been clear about the concept of organizational context. We define organizational context factors as dynamic and potentially modifiable. This is the main difference of organizational context variables from structural variables. Nevertheless, we will assess structural and contextual factors. We have added this to the introduction on p.2 lines 96-100 and clarified this in the data items section on p.5 line 210-212 and p.5 line 218-220.
Here, staffing was considered as an organizational context factor.	Thank you for mentioning this point. Staffing can be assumed to be one of many contextual factors. However, as described in the manuscript, focusing exclusively on staffing without taking other contextual factors into account is insufficient. We explained that on p.2 line 100-102.
I do not agree with this sentence. I read the paragraph above more than two times. I could just find the association between organizational context factors and patient outcomes. Please provide more rationales on the complicated interactions among	Thank you again for raising this important issue. We agree with the reviewer and apologize for these inaccuracies. To be consistent with the issue, we have added results from studies that analyzed staffing, context, and quality of care on p.2 lines 87-

organizational context factors, nurse staffing, and patient outcomes.	90. In addition, we have revised our concluding sentence on p.3 lines 116-120.
Why is this research problem important? Why should you and your colleagues conduct this research topic? Why should we, readers, read your research papers?	We feel that we clearly outlined the research gap (what is the evidence on the interaction between staffing and context and on the influence of this interaction on LTC quality) and the need to synthesize the available literature on this issue in the introduction. We also highlighted that the current literature that just focuses on the association of staffing and quality is inconclusive and not suited to inform actions to improve LTC quality. Finally, we point out that there is reason to believe that staffing and context interact in complex ways and that better understanding these complex relationships (by synthesizing the available evidence) may lead to powerful avenues to improve LTC quality. At the very minimum our review will identify research gaps related to the interaction between staffing and context that can be addressed in future studies.
Why are you going to synthesize quantitative research evidence only? The interactions between nurse staffing and organizational context factors may be better addressed by qualitative research. Please provide the rationales on it so that readers can understand (be convinced of) your decision.	Thank you, we appreciate the opportunity to clarify our understanding of interactions. We use the term 'interactions' as statistical interactions, not people talking with each other. We have clarified this on p.3 line 139-143 and defined statistical interactions in the inclusion criteria.
Why did not you include articles that addressed "how these interactions are associated with resident outcomes in long-term care"?	Thank you for this comment. This is exactly our research question (p.3 lines 123-129) and is covered by the inclusion criteria. Therefore, we include all articles that addresses this question.
Please provide exclusion criteria separately.	Thank you for mentioning this. We added the exclusion criteria on p.4 lines 154-163.
Cochrane Library and Web of Science—why did not you include those databases that are considered as the basic searching engine?	We appreciate the opportunity to clarify the selected databases. Cochrane Library and Web of Science were not included because: 1) Cochrane Reviews are indexed in Medline and the CENTRAL (RCTs) database actually identifies records from Medline, EMBASE and CINAHL to add to Cochrane's Centralised Search Service (CSS) (ie there is considerable overlap so it is unlikely the omitting this database will miss important literature); 2) For health related topics, the indexing coverage available for Web of Science is comparable to the indexing available in Scopus

Researchers cannot understand all types of languages. To defend it, please provide the details about the exclusion criteria.	Thank you for this advice and for the opportunity to further clarify the international background of our team and previous successful realizations of this strategy (p.3 lines 148-154).
Many articles related to your research topic have been published in various scholarly journals beyond nursing science. Please do not limit to the key journals you mentioned in order to enhance the scientific merits of your work. Also, the key authors were mainly trainees of the leading nursing workforce research groups. They shared the same perspective to the research problem (i.e., nursing shortage), and their research styles are also very similar each other. Please do not overconfident that mainstream nursing workforce research represents the truth about the research phenomenon. If you and your colleagues shows the willingness or the intention to reflect various perspectives and research styles	We agree with the reviewer and appreciate the opportunity to clarify the selection of key journals. We mention the key journals as an addition to the comprehensive database search. We will first search databases comprehensively, not limited to any journals, and then search selected journals by hand. This is good, common practice, and it is not possible to apply this additional hand search to a larger number of journals.
Good. Please provide magnitude guidelines to help decision making for you and your colleague and cite the relevant references.	Thank you for this advice. We added the relevant literature on p.4 line 188.
Please consider my previous comments I mentioned above and revise this sentence.	Thank you for this comment. Unfortunately, we did not understand this comment and would like ask for clarification please.
Please consider my previous comments I mentioned above and revise this sentence. "Modifiable" — do the structural variables have the only one difference point from organizational context? I cannot agree with your decision that you excluded the the structural variables and will only address organizational context. Please provide more robust, strong rationales on why organizational context should be more importantly addressed, compared to the the structural variables.	Thank you for this comment. Please see our response to your question above: *How different is "organizational context" from the structural variables?*
Please double-check grammar errors throughout the paper.	Thank you for this comment. We addressed this throughout the manuscript.
Please provide examples that can help readers' easier understanding of each concept.	Thank you for this advice. We added examples of each concept of organizational context for a better understanding (p.5 lines 215-219).
I would recommend you a re-categorization of these outcomes indicators. For example, (care process measures, care outcome measures, and utilization measures), or (quality of care	Thank you for your suggestion. However, we respectfully disagree with the reviewer here. We have followed the Donabedian model and added this information to the data items

outcome measures and utilization measures). Also, provide the rationales to support that those indicators are relevant and valid to LTC and cite them.	section (p.5 lines 220-226). We have also outlined the reasons why these indicators are relevant and valid in the context of long-term care (p.6 lines 227-229).
Please provide the relevant rationale(s) such as a framework or a guideline to support your data extraction.	Thank you for the opportunity to clarify the data extraction. We have added references from previous successful studies (p.6 lines 272-273).
i.e., i.e., ie, i.e., Be consistent throughout the paper, please.	Thank you for pointing this out. We addressed it throughout the paper
To verify non-significant statistical heterogeneity among included studies, Please try to state clearly, given that not all readers know meta-analysis.	Thank you for this correction, which we addressed on p.7 line 310.

Reviewer 2

Comment	Response
According to the policies of the BMJ Open the dates of the study should be included in the manuscript (start of the registration; the first search strategy; the planned date to conducted the search strategy and data extraction)	Thank you very much for this note. We have covered the additions within the other comments.

VERSION 2 – REVIEW

REVIEWER	Park, Claire Center for Econometric Optimization in the Nursing Workforce
REVIEW RETURNED	27-May-2022

GENERAL COMMENTS	1 Review of the manuscript “More than just staffing? Assessing evidence on the complex interplay among nurse staffing, other features of organizational context and resident outcomes in long-term care - a systematic review protocol” for the BMJ Open Dear Editor,

	Thank you so much for giving me one more opportunity to review the manuscript entitled “More than just staffing? Assessing evidence on the complex interplay among nurse staffing, other features of organizational context and resident outcomes in long-term care - a systematic review protocol.” This manuscript perfectly falls within my specialization and research domain. I very much enjoyed reviewing this manuscript. This exciting experience inspired me to contemplate the scholarly identity, vision, and promise of my own program of research. As a peer reviewer, I have also benefited from such a well-delineated advanced scientific article. I am truly honored beyond all measure to be a part of the panel of reviewers for the BMJ Open, and again, thank you for allowing me to review such a well-developed manuscript. I carefully double-checked the manuscript and confirmed that authors sufficiently addressed concerns and issues I raised in the previous review. I am satisfied with this revision, and I confirm you all that I have no more comments/queries. I humbly hope that my input was of help to both the BMJ Open and authors. To the authors, Thank you for your hard work! Warmest regards, Claire Su-Yeon Park